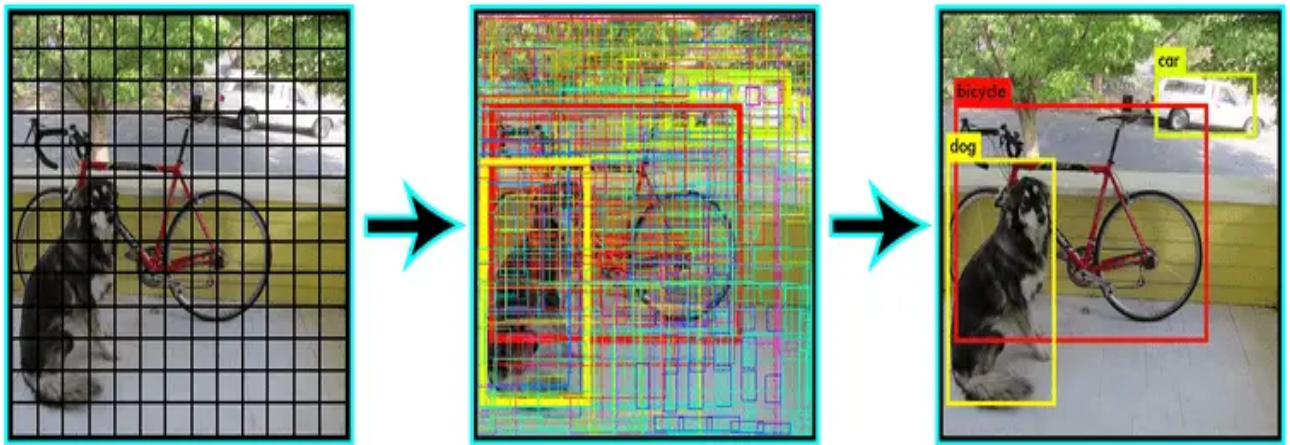




What Is Object Detection?

Object detection is a core problem in computer vision, it refers to the task of identifying *what objects* are in an image, and *where* they are. Unlike image classification (which simply says “this image contains a cat”), object detection draws **bounding boxes around each object and labels them**.

Core Concepts

At its heart, object detection does two things:

- **Classification** – recognizing what an object *is* (dog, bicycle, person, etc.)
- **Localization** – figuring out *where* it is in the image by predicting bounding box coordinates (x, y, width, height)

These two together form the backbone of detection systems.

Earlier classical methods like sliding windows and Haar cascades existed, but true performance came with convolutional neural networks (CNNs). Modern detectors like **R-CNN family**, **SSD**, and **YOLO** treat detection as a regression and classification problem solved by deep networks.

YOLO (“*You Only Look Once*”) is especially popular because it predicts bounding boxes and class probabilities in **one forward pass through a network**, making it real-time capable. We are going to use YOLO v5 for this project.

Steps in a Deep Learning Object Detection Workflow

1. Data Preparation

Gather images and annotate them with bounding boxes and class labels.

Tools like Labelling or Roboflow help annotate and export to formats like COCO or Pascal VOC.

2. Choose or Build Your Model

You can either:

- Use a **pre-trained model** (e.g., YOLOv5, YOLOv8)
- Train your own detector on custom data

Most videos teach how to fine-tune pre-trained models for your own objects.

3. Train the Model

Training involves feeding many images into the network, allowing it to learn features and bounding boxes. Typical training steps include:

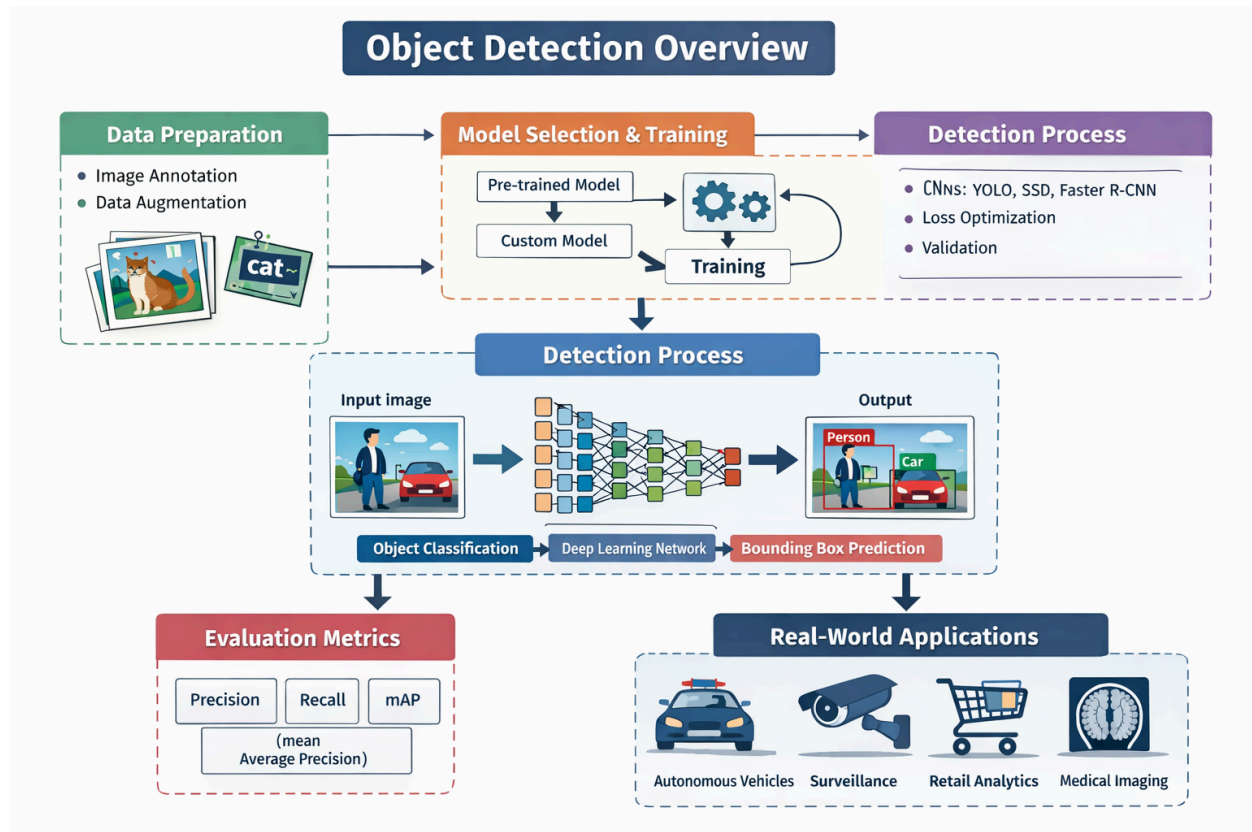
- Data augmentation (flip, crop, scale)
- Loss optimization
- Validation on held-out data

4. Evaluate Performance

Metrics such as **precision, recall, and mAP (mean Average Precision)** help quantify how good your detector is.

5. Deploy & Inference

Once trained, your model can run on live images, video streams, or embedded devices — often drawing boxes around detected objects.



Intuition behind working of the model

A neural network processes an image like looking through multiple lenses:

- Early layers detect simple features (edges, textures, patterns)
- Middle layers detect more complex patterns (wheels, faces, structures)
- Later layers detect full objects and their context

Bounding box prediction layers learn where features typically occur which allows the model to say "there's a person here, and a car there."

Real-World Uses of Object Detection

Object detection isn't just a Machine Learning concept it is widely used in day-to-day life

- **Autonomous vehicles** detect pedestrians and traffic signs.
- **Surveillance systems** count people and alert on suspicious activity.
- **Retail analytics** track product placement and customer movement.

- **AR/VR applications** overlay information on real-world objects.
- **Medical imaging** highlights anomalies in scans.

This cross-domain applicability is why object detection is such a hot skill in computer vision today.